

A Profit Optimization Model for Renewable Energy Communities based on the Distribution of Participants

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Abstract—After the growth of decentralized system in the energy sector due to many benefits, a new concept of Renewable energy community (REC) in Europe is introduced. REC, which comprises local generation of renewable energy sources regulated by citizens, enterprises and Public Administrations, has emerged because of the growth of renewable sources in European Union countries. There are many benefits, including social, environmental and economical for making RECs and becoming part of it. However, this concept of RECs as per RED II has presented researchers with numerous research topics and opportunities. Among these, self-consumption, energy sold and energy sharing stand out as key concepts, enabling the optimization of renewable energy sources within these RECs. This work focuses on the mathematical model considering the optimization in REC to maximize profit based on the number of prosumers and consumers. In the model, the consumers could be part of REC and also could change their role from consumer to prosumer following the constraints set by regulatory. The optimization model is developed and analyzed on hourly basis as per Italian framework. Furthermore, this work also focuses on the economic analysis using different techniques to find the profitability at different cases for the optimal REC and without REC. Numerical results are obtained for different seasons and show that there is an optimal distribution of prosumers and consumers in which maximum profit can be obtained.

Keywords—REC, Self-consumed Energy, Energy Sold, Energy shared, Optimization and Profitability

I. INTRODUCTION

Renewable energy sources (RESs) have become dominant due to their many advantages like high potential, environmental friendliness, cost-effective as compared to non-renewable energy sources in many countries depending upon the type of sources [1][2]. The integration of increasing amounts of RESs has risen due to the paradigm shift in electric power system in recent years [3][4]. As a result of this shift, the decentralized RESs are becoming more prevalent. Local control offered by decentralized energy systems makes community-based energy production and delivery an optimal solution [5]. By making use of the community's decentralized energy resources, these systems seek to meet the local energy need. In comparison to centralized energy systems, local decentralized energy resources minimize complexity, expense and inefficiency by bringing energy generation closer to users. They also strengthen local resilience, encourage energy independence and aid in the shift to zero carbon emissions [6]. Decentralized energy systems can spur creativity, give people more authority, and promote community involvement. Consequently, cooperative solutions known as

Energy Communities (ECs) have surfaced to enable the sharing of RES inside decentralized energy systems. These ECs share the objective of reducing energy use and encouraging active consumers to use energy more flexibly, which will lessen the heavy energy loads on the power grid [7]. An effective way to handle the change in the local energy landscape seems to be integrating local Distributed Renewable Energy sources (DRE) and involving local ECs [8]. Due to critical role of energy sector for the climate problem, the European Union unveiled the Clean Energy package, emphasizing that by 2030, 32% of the energy mix should come from RESs, and that the energy market should be modernized to take flexibility into consideration [9].

The EC has advanced significantly in Europe and is positioned to play a key role in building a more flexible and decentralized Energy Union where individuals are more influenced [10]. Having the inclusion in the Clean Energy Package and, notably, the 2018/2001 Renewable Energy Directive (RED II) introduced the concept of Renewable Energy Communities (RECs) and its creation focusing on the RESs usage while 2019/944 ED directives present the concept of CECs which focuses on the electricity having the combined primary aim delivering the social, economic, and environmental benefits to their members [11][12]. RECs and jointly acting RE self-consumers seem as new actors in the electricity sector as they provide a plethora of opportunities to promote community members' active participation and individual houses as the communities' fundamental units [13]. These ECs are in a good position to recognize regional energy requirements and bring people together around common goals like sustainability, self-sufficiency, resilience and autonomy [10]. Furthermore, local communities' roles change as they go from being simple consumers to prosumers who actively participate in demand response tactics, energy sharing, local generation, and energy efficiency initiatives [14].

One of the main challenges to be faced for the spread of RECs is represented by the research of the best configuration. Many Researchers are engaged on this emerging topic with various directions. F. Ceglia et al. [15] worked on the analysis of energy sharing directives, management analysis of old and current energy sharing directives and also environmental and economic perspectives. It was seen that RECs permit to avoid 39.5 t/y of CO₂ emissions in EU countries comparing the conventional design with the local self-consumption index of RES in EC. C. Gallego-Castillo et al. [16] carry out a regional analysis of the optimal self-consumption installations as per Spanish regulatory framework. The best installation size, both with and without compensation for

excess energy, results in cost reductions for self-consumers throughout the entire region, according to the findings V. Casalicchio et al. [17] worked on RECs Model focusing on the operation, investment and optimal Demand Side Management (DSM) and also on the evaluation of a fairness index in order to find the best allocation of benefits. In [18], the work models various design typologies of ECs in European countries like Portugal taking into account the various consumer types having heterogeneous profiles of electricity demand and willingness for participating, as well as multiple scenarios of technology deployment and electricity trading like peer-to-peer versus collective self-consumption. Moreover, research activity is going on, focusing on the optimization techniques for RECs with different energy sources (PV, Wind, ESS) and sizing, also including the economic factors, geographical site, management and operation. The aim is to focus on environment and economic aspects and other purposes like investment option, energy management system and energy sharing [19]-[25]. Our optimization model focuses on a different and original objective compared to previous research works, evaluating the best number distribution of prosumers and consumers inside the REC, in order to obtain the maximum profitability. The optimum distribution of the REC should fulfill the constraints set by the directive. The calculations were made for self-consumption, sold and shared energies and an optimal profit distribution of the number of prosumers and consumers has been evaluated. Results show that there is a configuration of the distribution of prosumers and consumers which maximizes the profit and the change of their role from consumer to prosumer is beneficial.

The work is organized in different sections: Section II focuses on the workflow and mathematical modeling in order to find the optimum number of participants and compare the profitability with and without REC. Section III discusses results obtained from the numerical simulations for different seasons and presents comparisons of results with and without optimization and In Section IV a discussion of the overall work is presented along with its perspectives.

II. MATHEMATICAL FORMULATION

The mathematical model considers a number of prosumers ($1, 2, 3, \dots, N_p$) and consumers ($1, 2, 3, \dots, N_c$) having electrical load as P_L and C_L , respectively. The REC is built by referring to a primary substation subtend to a geographical area. The maximum power admitted for the REC is 1 MW as per the Italian regulatory framework [26]. In the numerical model the total number of participants is considered fixed, but the distribution of prosumers and consumers can change, letting a consumer become a prosumer and vice versa. This work has the main aim to find and check profitability based on the optimal distribution of consumers and prosumers. The load of prosumer can be partially or totally satisfied by the Photovoltaic (PV) production and it represents the Self-consumption. In the case of excess of PV generation, the excess power can be supplied to the grid, which is considered as sold energy and if consumers require energy in that time (hourly basis), the minimum between the excess energy and the load energy is considered as shared energy. A sketch of the model is shown in Fig. 1 based on the number of consumers and prosumers. The data for PV production for the prosumers is

taken from PVGIS [27] and PV production could be individual for the prosumers or by means of a common plant depicted in the figure. The load for both prosumers and consumers are also used from data given by Ricerca sul Sistema Energetico (RSE) [28].

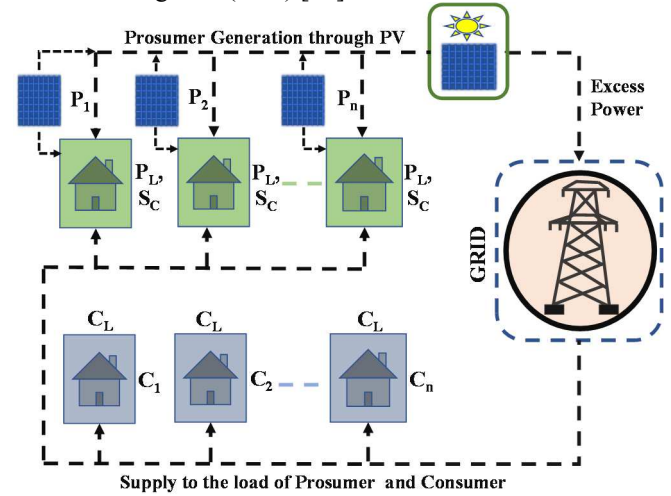


Fig. 1. REC Model

The model includes the constraints on the maximum power of the REC and on the total number of participants N_t , but it allows the shift between participants who can change their role. Consumers could become prosumers (from t_0 to t_1 , Fig. 2) if they want to join following the limits of REC as capacity, which should not be increased beyond the limit of 1 MW. This is also shown in Fig. 2, the consumers could become prosumers from t_1 .

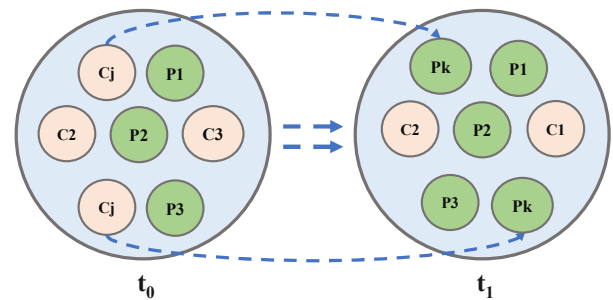


Fig. 2. Sketch of the REC in which consumer C_j at t_0 can become prosumer P_k at t_1 .

Profitability could be increased by considering that in REC the excess power can be sold to the grid and at the same time shared among the REC members with a supplementary economic incentive. Selling energy to the grid is not a common or unique concept, however sharing of energy has become unique because of the economic incentives given to REC members like prosumers and consumers. Consumers can use energy at the same price but inside a REC they receive a part of the benefit of the incentive and also for prosumers the additional benefit is represented by the incentive due to the shared energy and the reduced transport cost of energy. This encourages local people to participate in the REC and take benefits by utilizing the RESs power. Moreover, REC has further benefits like social and environmental. The mathematical model is illustrated in the flowchart captioned in Fig. 3.

Calculations start from a combination of the number of prosumers and consumers, and when all the possible combinations are considered the maximum profit can be carried out along with its prosumer and consumer distribution.

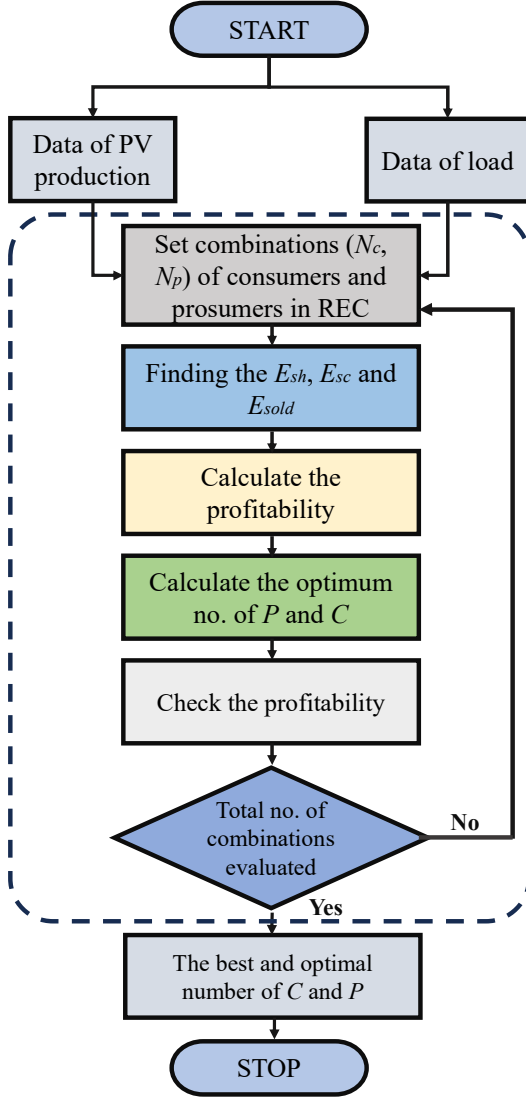


Fig. 3. Overall mathematical and model workflow

By assuming that all the prosumers have the same energy demand (D_p), it is possible to design, for each prosumer, the size of the plant considering that the PV power of a single prosumer is evaluated in order to satisfy with its producibility a fixed percentage of the load and the following equation is used:

$$P_n = \frac{S\% \cdot D_p}{E_{PV}} \quad (1)$$

where

- P_n is nominal power of prosumer's plant in kWp.
- D_p is the yearly prosumer's demand in MWh.
- $S\%$ is the percentage of satisfaction of yearly prosumer's demand.
- E_{PV} is the yearly production of energy for 1 kW plant [MWh/kWp].

For all the REC, hourly shared energy ($E_{sh,h}$) it is defined as the minimum value among the energy produced by the

photovoltaic plant, which isn't self-consumed by prosumers (excess energy which is sold to the grid) and total hourly energy demand ($D_{E,h}$):

$$E_{sh,h} = \min(E_{sold,h}; D_{E,h}) \quad (2)$$

$$D_{E,h} = \sum_{i=1}^{N_p} L_{res,i,h} + \sum_{j=1}^{N_c} C_{L,j,h} \quad (3)$$

where

- $E_{sold,h}$ is the excess energy produced by the photovoltaic plant computed as follows:

$$E_{sold,h} = \max(\sum_{i=1}^{N_p} (PPV_{i,h} - E_{self,i,h}); 0) \quad (4)$$

in which

- $E_{self,i,h}$ is the energy produced by the photovoltaic plant self-consumed by the prosumer i

$$E_{self,i,h} = \min(PPV_{i,h}; P_{L,i,h}) \quad (5)$$

- $C_{L,j,h}$ is the hourly energy load of prosumer j ;
- $PPV_{i,h}$ is the hourly energy production of the plant owned by prosumer i .
- $L_{res,i,h}$ is the amount of $P_{L,i,h}$, which it is not satisfied by $PPV_{i,h}$, it can be computed as it follows:

$$L_{res,i,h} = \max(\sum_{i=1}^{N_p} (P_{L,i,h} - PPV_{i,h}); 0) \quad (6)$$

If equation (2) is combined with (4), (5) and (6), it is possible to obtain (7):

$$E_{sh,h} = \min(\max(X; 0); \sum_{j=1}^{N_c} C_{L,j,h} - \min(X; 0)) \quad (7)$$

$$X = \sum_{i=1}^{N_p} PPV_{i,h} - P_{L,i,h} \quad (8)$$

According to European and Italian law, it is necessary that, constraint in equation (9) must be satisfied:

$$N_p \cdot P_n \leq 1 \text{ MW} \quad (9)$$

The calculation of the energy flows represents the input data for the profitability of REC using the Net Present Value method evaluated with the following expression [29]

$$NPV = \sum_{n=0}^t \frac{C_f(t)}{(1+r)^t} \cdot \frac{1}{(1+i)^t} \quad (10)$$

$$C_{f,o} = \sum_{n=0}^t [CX + OX + RX]_n \quad (11)$$

$$C_{f,i} = \sum_{n=0}^t [C_{E_{sc}} + C_{E_s} + C_{E_{sh}}]_n \quad (12)$$

where

- C_f is the total cashflows
- $C_{f,o}$ are the cash outflows
- $C_{f,i}$ are the cash inflows
- CX and OX are CAPEX and OPEX
- RX is the rest of costs
- $C_{E_{sc}}$ is self-consumed energy cost
- C_{E_s} is the sold energy cost
- $C_{E_{sh}}$ is the shared energy cost
- i is the decay in percentage

- r is the discount rate
- t is the lifetime [year]

Continuing with the above, the Profitability could be found using the NPV value and investment. The profitability has been computed focusing on the parameters with and without REC to check and compare the results. The equations are given as:

$$\Delta_{Profit} = P_{REC} - P_{No REC} \quad (13)$$

Here,

$$\Delta_{Profit} = C_{sh} \cdot E_{sh} + C_d \cdot E_{sc} - C_{man} \quad (14)$$

where

- P_{REC} is the profit considered in the case of REC
- P_{NoREC} is the profit considered without REC
- Δ_{Profit} term is used to see the profit by comparing with and without REC
- C_{sh} is the amount of price given by Gestore dei Servizi Energetici [26] as valorization of shared energy in a REC, according to Italian law (Decreto CER [30])
- C_d is the amount of price given by GSE (as valorization of distribution power loss avoided by E_{self})
- C_m is the management cost

The profitability for the REC could be calculated using equations (10) to (14). The data used to find the profitability using the above equations is given in Table II and some of them taken from [31] [32].

TABLE II Technical and Economical Specifications

Different technical and economical specifications and cost details			
S No.	Details/Description	Value	Unit
1	PV Unitary cost(CX)	1100	€/kWp
2	Design cost (CX)	200	€/kWp
3	Insurance cost	0.50	% of CX
4	ICT & Metering infrastructure unitary cost	200	€/kWp
5	advocate fees for agreement/management fees ($C_{management}$)	500	€/y
6	OX	2.5	% of CX
7	OX growth /inflation rate	1	%
8	PV Performance decay(i)	1	% (/y)
9	Project lifespan	20	years
10	Discount rate	3	%
11	C_{sh}	0.12	€/kWh
12	C_d	0.00848	€/kWh

III. RESULTS AND ANALYSIS

Numerical results are presented for a closed ($N_p + N_c = N_t = 100$) REC using photovoltaic (PV) system, located in Potenza (Italy) [33], with the possibility for consumers to become prosumers. Three different sets of simulations were made by assuming different values of S% (60 %; 80 %; 95 %), in order to find if there is an optimal configuration in terms of profit.

Both prosumers and consumers have the same load profile, which was calculated by using data given by RSE, considering a single-family house with two floors. Data have been calculated on a model that takes as input data, the building's characteristics and the socio-demographic variables present within a data collection carried out by ISTAT, the result of a survey conducted on a sample of 20'000 Italian families.

The target of this analysis is to identify the best REC configuration, in order to maximize Δ_{Profit} , as given by equation (14).

The optimum result is reached with 40 prosumers and 60 consumers, as it shown in the Fig. 4 if the theoretical percentage of satisfaction of yearly prosumer's demand is equal to 80% or 95%, that means that the total installed nominal power is 50 kWp or 59.3 kWp. If the percentage is equal to 60 % the optimal configuration is $N_p = 50$ with 46.8kWp. Numerical results show that there is an optimal configuration for the REC obtained with a particular combination of the number of prosumer and consumer and this result could be relevant in the design and future management of the REC.

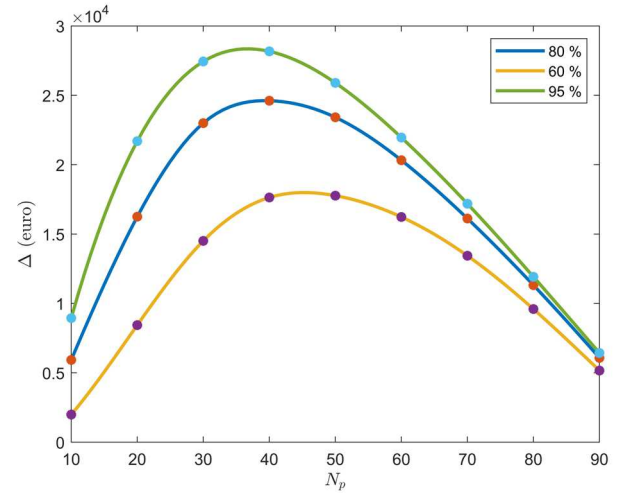


Fig. 4. Δ_{Profit} as function of N_p , if $N_p + N_c = 100$

In Figures 5-8, the shared energy, sold energy, the sum of consumer's loads and of the residual loads are shown for three days during each season of the year for a REC with 40 prosumers and 60 consumers, which is the optimal REC if the theoretical percentage of satisfaction of yearly prosumer's demand is 80%. In Fig. 7, there is a high peak value of REC load, this could be observed possibly in summer due to the higher energy demand from the REC due to cooling system of REC's buildings.

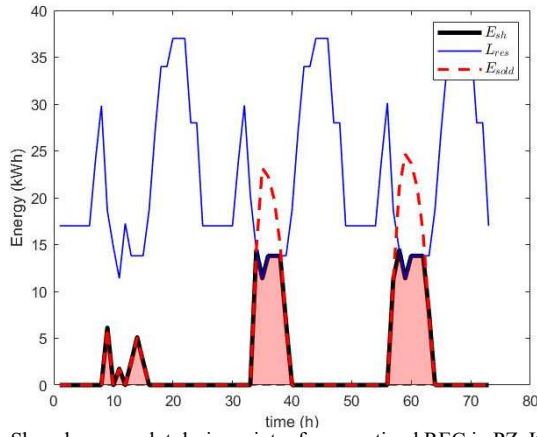


Fig. 5. Shared energy plot during winter for an optimal REC in PZ, Italy

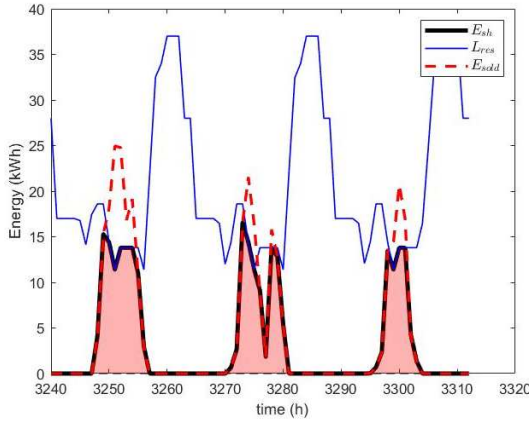


Fig. 6. Shared energy plot during spring for an optimal REC in PZ, Italy

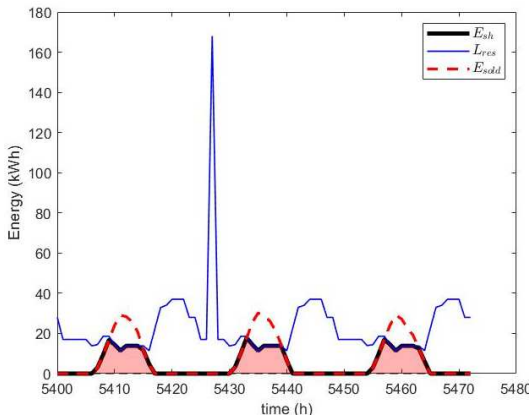


Fig.7. Shared energy plot during summer for an optimal REC in PZ, Italy

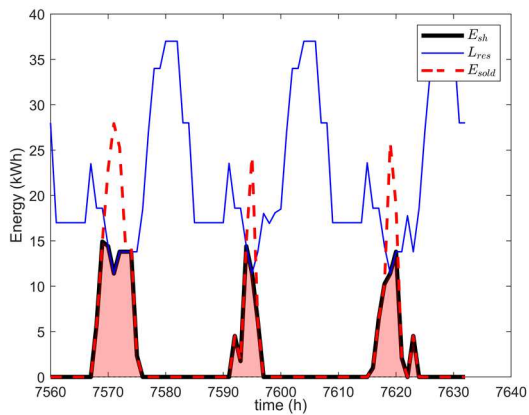


Fig.8. Shared energy plot during autumn for an optimal REC in PZ, Italy

In Figure 9 sold energy and the sum of consumer's loads and of the residual loads of the prosumers are shown for three days during autumn season of the year for a REC with 90 prosumers and 10 consumers ($S\% = 80\%$, $P_n = 112.4$ kWp) and for a REC with 10 prosumers and 90 consumers ($S\% = 80\%$, $P_n = 12.5$ kWp). By comparing autumn results for optimal REC reported in Fig. 8 with the results in Fig. 9 it is possible to see that in optimal REC a greater amount of excess energy is shared inside the community. In the non-optimal condition (Fig. 9) there is a small amount of shared energy, which generates a lower profit. The other border case is reported in Fig. 10, in which it is possible to see that in a REC with 10 prosumers and 90 consumers there is not enough surplus of energy produced by the plant in order to satisfy REC's energy demand or to sell it to the grid.

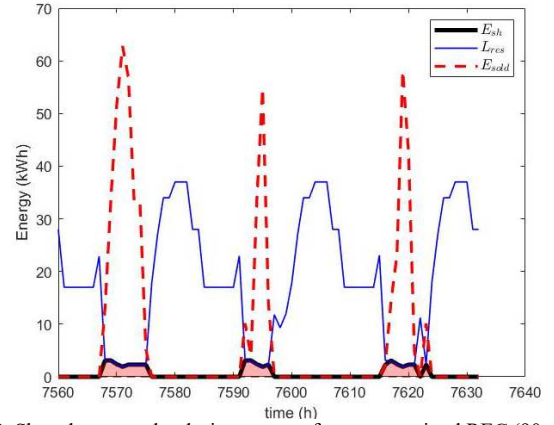


Fig. 9. Shared energy plot during autumn for a non-optimal REC (90 prosumers and 10 consumers)

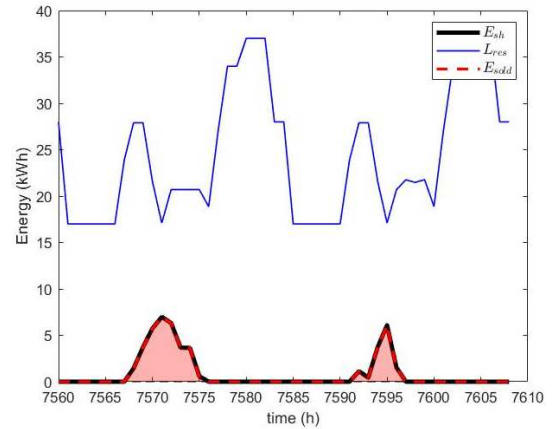


Fig.10. Shared energy plot during autumn for a non-optimal REC (10 prosumers and 90 consumers)

IV. CONCLUSION

The RECs, as per RED II directive, has become an emerging topic in Europe due to many advantages like social, economic, and environmental. This topic has engaged many stakeholders, researchers, enterprises, citizens, and public administration. RECs, which include local generation of RESs like solar, wind, biogas, hydro, are regulated by citizens and have emerged because of the rise of citizen-owned production units. Among these, self-consumption, sold and shared energy stand out as key concepts due to incentives and profits given to their participants.

Our work is focused on the mathematical model considering the optimization in REC using photovoltaic (PV) system to maximize profit based on the number of prosumers and

consumers. The consumers at t_0 could be part of REC and could become prosumers at t_1 following the constraints. The optimization model is analyzed on an hourly basis showing the positive outcomes. An optimum distribution of prosumers and consumers have been calculated for different level of self-consumption, showing that the design stage can be improved during the life of the REC. Energy fluxes have been evaluated on hourly basis in a proper economic model by using different techniques to find the profitability as the gain of the REC respect to the case of single prosumers and consumers (i.e., the case of absence of the REC). Results are presented for each season and comparisons have been reported between REC and not REC case.

This work could be a helpful tool for the design and management of a REC and can be extended in future in order to consider different classes of renewable production and load in which participant could be inserted. Class distribution could give a more detailed analysis of information to be adopted in the design optimization of a REC. Moreover, in the study the role of storage can be included in order to investigate the possible technical and economical benefit on the profit.

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